

Homework 5

Due: 26th February, 2026

Exercise 5.1 (Panaretos, Exercise 25). Let $\{X_n\}_{n=1}^\infty$ be a sequence of random variables defined by

$$X_n := (-1)^n X$$

where X is a Rademacher random variable

$$\mathbb{P}(X = 1) = \frac{1}{2}, \quad \mathbb{P}(X = -1) = \frac{1}{2}.$$

1. Compute the cumulative distribution function F_{X_n} of X_n .
2. Show that $X_n \xrightarrow{d} X$.
3. Show that the sequence $\{X_n\}$ **does not** converge in probability to X . *Hint: if n is odd, what is $|X_n - X|$?*

Sol. First, I'm actually going to compute the cdf of X . Notice that X only takes two values: -1 and $+1$. So, for any $x < -1$ we have $\mathbb{P}(X \leq x) = 0$. For any x such that $-1 \leq x < 1$, we just have $\mathbb{P}(X \leq x) = \mathbb{P}(X = -1) = \frac{1}{2}$ and for any $x \geq 1$, $\mathbb{P}(X \leq x) = 1$. In summary,

$$F_X(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{1}{2} & \text{if } -1 \leq x < 1 \\ 1 & \text{if } x \geq 1. \end{cases} \quad (1)$$

Now, let us consider X_n . If n is even, $(-1)^n = 1$ and so $X_n = (-1)^n X = X$. Then, $F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$. Otherwise, $(-1)^n = -1$ and $X_n = -X$. There, X_n also only takes two values:

$$X_n = 1 \iff X = -1 \implies \mathbb{P}(X_n = 1) = \mathbb{P}(X = -1) = \frac{1}{2}$$

and

$$X_n = -1 \iff X = 1 \implies \mathbb{P}(X_n = -1) = \mathbb{P}(X = 1) = \frac{1}{2}.$$

Notice that the probability mass function of X_n is the same as X , so even though X_n is not equal to X we still will have by the same computation that

$$F_{X_n}(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{1}{2} & \text{if } -1 \leq x < 1 \\ 1 & \text{if } x \geq 1. \end{cases}$$

which agrees with (1). So, for all x and for all n , $F_{X_n}(x) = F_X(x)$ and hence $\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$. Thus, $X_n \xrightarrow{d} X$, as desired.

However, the convergence does not hold in probability. For all odd n , $X_n = -X$ and so $X_n - X = -2X$. Since $X \in \{\pm 1\}$, $|-2X| = 2$ always, and so $|X_n - X| = 2$ for all odd n . In particular, for all odd n ,

$$\mathbb{P}(|X_n - X| \geq 1) = 1$$

and so it is impossible to have $\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq 1) = 0$. However, since for convergence in probability to hold we need $\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \epsilon) = 0$ for all $\epsilon > 0$, we see that X_n does not converge in probability to X . The difference here is in actual sample values versus distribution: for all odd n , the sample values of X_n never agree with those of X , but the sampling distribution of X_n always agrees with the sampling distribution of X (i.e. they would have similar histograms if we took a large number of samples) for all n .

Exercise 5.2. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$. Define the estimator $\hat{\lambda} := \frac{1}{n} \sum_{i=1}^n X_i$, and recall that $\hat{\lambda}$ is sufficient for λ .

1. Compute the bias, $\text{bias}(\hat{\lambda}, \lambda)$.
2. Determine the mean-squared error $\text{MSE}(\hat{\lambda}, \lambda)$.
3. Show that $\hat{\lambda}$ is consistent for λ , either by appeal to the law of large numbers or our MSE inequality.

Sol. For the bias, we have

$$\text{bias}(\hat{\lambda}, \lambda) = \mathbb{E}[\hat{\lambda}] - \lambda = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] - \lambda = \frac{1}{n} \sum_{i=1}^n \lambda - \lambda = \frac{n\lambda}{n} - \lambda = \boxed{0}.$$

Hence, the mean-squared error is

$$\text{MSE}(\hat{\lambda}, \lambda) = \text{bias}(\hat{\lambda}, \lambda)^2 + \text{Var}(\hat{\lambda}) = 0 + \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \lambda = \frac{n\lambda}{n^2} = \boxed{\frac{\lambda}{n}}$$

using that the X_i are i.i.d. and that $\text{Var}(\text{Poisson}(\lambda)) = \lambda$. Finally, since $\mathbb{E}[\text{Poisson}(\lambda)] = \lambda$, the law of large numbers tells us that

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \lambda$$

and hence $\hat{\lambda}$ is consistent for λ , as desired.

Exercise 5.3. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Unif}(0, \theta)$, and recall that $\hat{\theta} = \max(X_1, \dots, X_n)$ is sufficient for θ .

1. Compute the cumulative distribution function of $\hat{\theta}$.
2. Use the previous answer to find the probability density function of $\hat{\theta}$.
3. Use your answer to the previous part to find $\mathbb{E}[\hat{\theta}]$ and then the bias, $\text{bias}(\hat{\theta}, \theta)$.
4. Compute $\text{Var}(\hat{\theta})$, and use this to find $\text{MSE}(\hat{\theta}, \theta)$.
5. Show that $\hat{\theta}$ is consistent for θ .

Sol. To find the cdf, we first write

$$F_{\hat{\theta}}(t) = \mathbb{P}(\hat{\theta} \leq t) = \mathbb{P}(\max(X_1, \dots, X_n) \leq t).$$

Now, $\max(X_1, \dots, X_n) \leq t$ if and only if $X_i \leq t$ for all $1 \leq i \leq n$, hence

$$\mathbb{P}(\hat{\theta} \leq t) = \mathbb{P}(X_1 \leq t, X_2 \leq t, \dots, X_n \leq t).$$

Since the samples are independent, we may write

$$\mathbb{P}(\hat{\theta} \leq t) = \prod_{i=1}^n \mathbb{P}(X_i \leq t) = F_{\text{Unif}(0, \theta)}(t)^n$$

since each $X_i \sim \text{Unif}(0, \theta)$. Now, the cumulative distribution function of a $\text{Unif}(0, \theta)$ random variable is given by

$$F_{\text{Unif}(0, \theta)}(t) = \begin{cases} 0 & \text{if } t \leq 0 \\ \frac{t}{\theta} & \text{if } 0 < t < \theta \\ 1 & \text{if } t \geq \theta \end{cases}$$

and so

$$F_{\hat{\theta}}(t) = \begin{cases} 0 & \text{if } t \leq 0 \\ \frac{t^n}{\theta^n} & \text{if } 0 < t < \theta \\ 1 & \text{if } t \geq \theta. \end{cases}$$

We can also differentiate to find the probability density function of $\hat{\theta}$ as

$$f_{\hat{\theta}}(t) = F'_{\hat{\theta}}(t) = \begin{cases} \frac{nt^{n-1}}{\theta^n} & \text{if } t \in (0, \theta) \\ 0 & \text{otherwise.} \end{cases}$$

Thus, its expected value is

$$\mathbb{E}[\hat{\theta}] = \int_0^\theta t \frac{nt^{n-1}}{\theta^n} dt = \frac{n}{\theta^n} \int_0^\theta t^n dt = \frac{n}{(n+1)\theta^n} t^{n+1} \Big|_0^\theta = \frac{n\theta^{n+1}}{(n+1)\theta^n} = \boxed{\frac{n}{n+1}\theta}.$$

The bias is

$$\text{bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta} - \theta] = \frac{n}{n+1}\theta - \theta = \boxed{-\frac{\theta}{n+1}}.$$

The second moment of $\hat{\theta}$ is

$$\int_0^\theta t^2 \frac{nt^{n-1}}{\theta^n} dt = \frac{n}{\theta^n} \int_0^\theta t^{n+1} dt = \frac{n}{(n+2)\theta^n} t^{n+2} \Big|_0^\theta = \frac{n\theta^{n+2}}{(n+2)\theta^n} = \frac{n\theta^2}{(n+2)}.$$

So, the variance can be computed as

$$\begin{aligned} \text{Var}(\hat{\theta}) &= \mathbb{E}[\hat{\theta}^2] - \mathbb{E}[\hat{\theta}]^2 = \frac{n\theta^2}{n+2} - \left(\frac{n\theta}{n+1}\right)^2 = \theta^2 \left(\frac{n}{n+2} - \frac{n^2}{(n+1)(n+1)} \right) \\ &= \theta^2 \left(\frac{n(n^2 + 2n + 1) - n^2(n+2)}{(n+2)(n+1)^2} \right) = \boxed{\frac{n\theta^2}{(n+2)(n+1)^2}}. \end{aligned}$$

We see that the mean-squared error is

$$\text{MSE}(\hat{\theta}, \theta) = \left(\frac{-\theta}{n+1}\right)^2 + \text{Var}(\hat{\theta}) = \frac{\theta^2}{(n+1)^2} + \frac{n\theta^2}{(n+2)(n+1)^2} = \frac{(2n+2)\theta^2}{(n+2)(n+1)^2} = \boxed{\frac{2\theta^2}{(n+2)(n+1)}}.$$

Notice that the mean-squared error tends to zero as $n \rightarrow \infty$. Importantly, we have for any $\epsilon > 0$ that

$$\mathbb{P}(|\hat{\theta} - \theta| \geq \epsilon) \leq \frac{\text{MSE}(\hat{\theta}, \theta)}{\epsilon^2} = \frac{2\theta^2}{(n+2)(n+1)\epsilon^2} \rightarrow 0$$

and so $\hat{\theta} \xrightarrow{P} \theta$ as $n \rightarrow \infty$. Thus, $\hat{\theta}$ is consistent for θ , as desired.

Notice that unlike in the Cramér-Rao lower bound, the denominator is order n^2 !